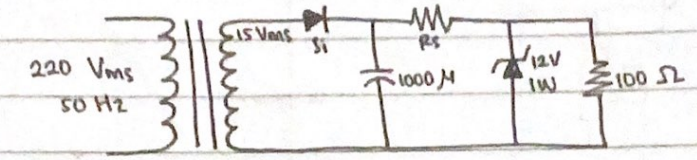
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1A - TNK
231349015

No:
Date:

Latihan Soal Dioda Zener, BJT dan FET

1. Pada rangkaian stabilizer tegangan pada gambar, tentukan :



a) Besarnya tegangan DC pada kapasitor

↳ $V_{s(p)} = V_{eff} \times \sqrt{2}$

$= 15 \times \sqrt{2}$

$= 21,21\ V$

$V_{s(p)DC} = V_{s(p)} - V_D$

$= 21,21 - 0,7$

$= 20,51\ V$

b) Arus maksimum pada dioda zener

↳ $I_{zmax} = \frac{P_{zmax}}{V_z} = \frac{1}{12} = 0,0833\ A = 83,3\ mA$

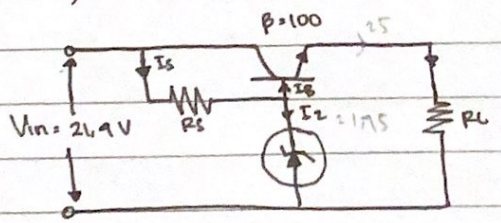
c) Nilai R_s terkecil supaya P_z max tidak terlampaui

↳ $P_z = I_{z\,max}^2 \times R_s$

$1 = (0,0833)^2 \times R_s$

$R_s = \frac{1}{0,00694} = 144,1\ \Omega$

2. Rangkaian stabilizer tegangan pada gambar menggunakan transistor dengan $\beta = 100$, $V_{BE} = 0,7\ V$ dan Dioda Zener 12 V. Bila $V_{in} = 21,9\ V$, $I_z = 1,75 mA$ dan diinginkan $I_L = 25 mA$. Tentukan :



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a) Nilai R_S

$$I_B = \frac{I_C}{\beta} = \frac{25\ mA}{100} = 0{,}25\ mA$$

$$I_S = I_Z + I_B$$

$$I_S = 1{,}75\ mA + 0{,}25\ mA$$

$$I_S = 2\ mA$$

$$R_S = \frac{V_{in} - V_{BE}}{I_S} = \frac{21{,}4 - 0{,}7}{2\ mA} = \frac{20{,}7}{2 \times 10^{-3}} = 10{,}35\ k\Omega$$

b) Besar tegangan pada R_L

$$V_L = V_Z - V_{BE}$$

$$= 12 - 0{,}7$$

$$= 11{,}3\ V$$

c) Besar tegangan V_{CE}

$$V_{CE} = V_C - V_E$$

$$= 21{,}4 - 11{,}3$$

$$= 10{,}1\ V$$

d) Dissipasi daya pada transistor

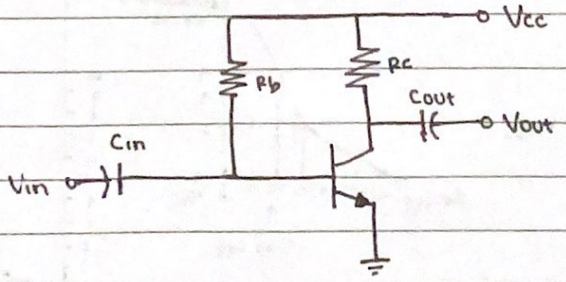
$$P_d = (V_{CE} \times I_C) + (V_{BE} \times I_B)$$

$$= (10{,}1 \times 25\ mA) + (0{,}7 \times 0{,}25\ mA)$$

$$= (10{,}1 \times 25 \times 10^{-3}) + (0{,}7 \times 0{,}25 \times 10^{-3})$$

$$= 0{,}253\ Watt$$

3. Pada gambar, bila $V_{CC} = 12V$, $R_B = 390k\Omega$, $R_C = 1{,}5k\Omega$, dan $\beta = 150$. Tentukan:



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a) I_B

$I_B = \frac{V_{CC} - V_{BE}}{R_B} = \frac{12 - 0,7}{390k} = 2,897 \times 10^{-5}$ A

b) I_C

$I_e = \beta \times I_B$

$= 150 \times (2,897 \times 10^{-5})$

$= 4,345 \times 10^{-3}$ A

c) V_{CE}

$V_{CE} = V_{CC} - (I_e \times R_c)$

$= 12 - (4,345 \times 10^{-3} \times 1500)$

$= 12 - 6,52$

$= 5,48$ V

d) PCE

$P_c = V_{CE} \times I_c$

$= 5,48 \times 4,345 \times 10^{-3}$

$= 0,024$

$P_E = V_E \times I_E$

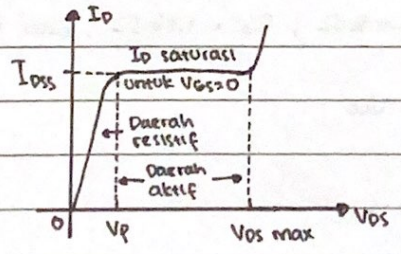
$= 0$

$P_{CE} = P_c - P_E$

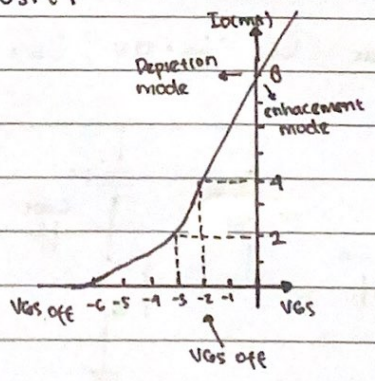
$= 0,024 - 0$

$= 0,024$ Watt

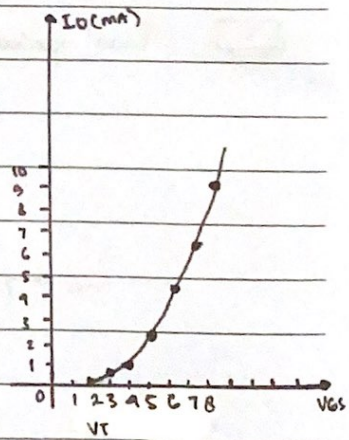
4. a) Gambarkan kurva karakteristik Transkonduktansi ID = (VGS) untuk JFET, D-MOSFET dan E-MOSFET



(JFET)



(D-Mosfet)

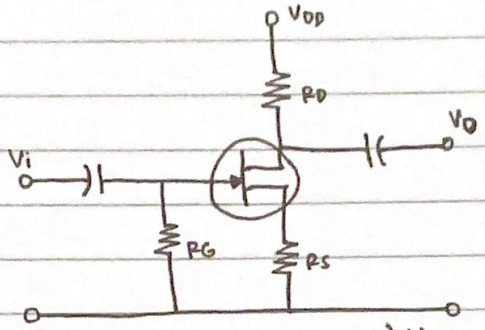


(E-Mosfet)

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b) Untuk rangkaian JFET, bila diinginkan $V_{GSQ} = 1/4\ V_{GSoff}$ dan $V_{DSQ} = 1/2\ V_{DD}$. Tentukan nilai R_S dan R_D bila parameter JFET sbb: $I_{DSS} = 8mA$; $V_{GSoff} = -4\ V$



•) $V_{GS} = V_G - V_S$

$-1 = 0 - V_S$

$V_S = 1\ V$

•) $I_D = I_{DSS}\left[1 - \frac{V_{GS}}{V_P}\right]^2$

$= 8mA\left[1 - \frac{-1}{-4}\right]^2$

$= 4.5mA$

•) $V_{DD} = I_D R_D + V_{DS} + I_D R_S$

•) $R_D = \frac{V_{DD} - V_{DS} - V_S}{I_D}$

$= \frac{12 - 6 - 1}{4.5mA}$

$= 1.1k\Omega$

•) $R_S = \frac{V_S}{I_D} = \frac{1}{4.5mA} = 222.2\Omega$

c) Tentukan pula besar R_G sehingga input rangkaian mempunyai tahanan input sebesar $1M\Omega$

↳ $R_G = 1M\Omega$